5. $\frac{(3x+2)}{6}\sqrt{9x^2+12x+3}-\frac{1}{6}\log\left|(3x+2)+\sqrt{9x^2+12x+3}\right|+c$

6. $\frac{1}{3}\left[(x+1)^3\log x-\frac{x^3}{3}-\frac{3x^2}{2}-3x-\log|x|\right]+c$

7. $x\log\left(x-\sqrt{x^2-1}\right)+\sqrt{x^2-1}+c$ 8. 0 9. $\frac{1}{4}\left[\frac{e^4-5}{e^2}\right]$ 10. $\frac{14}{15}$

3. Integral Calculus – II

Exercise 3.1

1. 5 sq.units 2. 2 sq.units 3. $\frac{8a^2}{3}sq.units$ 4. $\frac{3}{2}sq.units$

5. $\frac{17}{2}sq.units$ 6. $\frac{8}{3}sq.units$ 7. $\frac{32}{3}sq.units$

Exercise 3.2

1. ₹28,000 2. $y=\left(\frac{2x-1}{3x+2}\right)$ 3. $P=8-2x,\ R=8x-2x^2$

4. ₹4,419 5. ₹5,680

6. Total cost $=100x-5x^2+\frac{0.1x^3}{3}+500$, Average Cost $=100-5x+\frac{x^2}{30}+\frac{500}{x}$

7. Total Cost $=\frac{1500}{7}x^{\frac{7}{5}}$, Average Cost $=\frac{1500}{7}x^{\frac{2}{5}}$

8. Cost function $C=2\sqrt{ax+b}-2\sqrt{b}$ 9. ₹14,133.33

10. Total Revenue=₹5,95,000 11. Demand function $P=9-\frac{4x^2}{3}$

13. Demand function $P=20e^{-\frac{x}{10}}$ 14. Profit function $=13x-0.065x^2-120$

15. Revenue function $R=1500x-2x^2-x^3$, Average revenue function $P=1500-2x-x^2$

16. Revenue function $R=10x+\frac{3x^2}{2}-\frac{x^3}{3}$, Demand function $P=10+\frac{3x}{2}-\frac{x^2}{3}$

17. Total Cost $C=4000\sqrt{7x+4}+18000$, Average Cost $A.C=\frac{4000}{x}\sqrt{7x+4}+\frac{18000}{x}$

18. $C=\frac{x^2}{4}+5000$ 19. Revenue function $R=20x-\frac{5x^2}{2}+x^3$

20. Demand function $P=14-3x+3x^2$

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